

Technology Stack

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Technology Stack

A technology stack is a combination of programming languages, frameworks, libraries, development tools, and platforms used to design, develop, and deploy a software application. Selecting an appropriate technology stack ensures that the application is reliable, scalable, maintainable, and capable of delivering accurate results.

For the **Credit Card Approval Prediction using Machine Learning** project, technologies have been selected to support efficient data preprocessing, machine learning model development, web application development, version control, and deployment. The chosen tools provide a simple development environment while ensuring accurate prediction and smooth interaction between the frontend and backend.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	Used to create a responsive, user-friendly, and visually appealing web interface for collecting applicant details and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Python is used for implementing machine learning logic, while Flask provides a lightweight framework to connect the frontend with the trained model and process user requests efficiently.
3	Database / Data Storage	CSV Dataset, Joblib (.pkl Model File)	The application uses CSV files for training data and stores the trained Random Forest model as a .pkl file using Joblib. No relational database is required for the current version.
4	Cloud / Hosting / Deployment	Local Flask Server (Future: Render/Heroku/AWS)	The application is currently deployed on a local Flask development server for testing and demonstration. It can be deployed to cloud platforms in future versions for public access and scalability.

5	Version Control & CI/CD	Git, GitHub	Git is used for source code version control, while GitHub is used to store the project repository, manage changes, and support collaboration among team members.
6	Third-Party Libraries / Other Tools	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Joblib, VS Code, Jupyter Notebook	Pandas and NumPy are used for data preprocessing, Scikit-learn for building and evaluating the machine learning model, Matplotlib and Seaborn for data visualization, Joblib for model serialization, Jupyter Notebook for model development, and Visual Studio Code for application development.